PERGAMON

Journal of Geodynamics 28 (1999) 321–340

JOURNAL OF GEODYNAMICS

Paleomagnetism, geochemistry and $^{40}Ar/^{39}Ar$ dating of the North-eastern Paraná Magmatic Province: tectonic implications

M. Ernesto[a,*], M.I.B. Raposo[a,1], L.S. Marques[a], P.R. Renne[b], L.A. Diogo[a], A. de Min[c]

[a]*Instituto Astronômico e Geofisico, Universidade de São Paulo Rua do Matão,1226, Cidade Universitária, 05508-900, São Paulo, Brazil*

[b]*Berkeley Geochronology Center and Department of Geology and Geophysics, University of California, 2455 Ridge Road, Berkeley, CA, 94709 USA*

[c]*Dipartimento di Scienze della Terra, Universitá di Trieste Via E. Weiss 8, 34127, Trieste, Italy*

Received 10 September 1998; received in revised form 1 February 1999; accepted 27 March 1999

Abstract

Seventy sites of sills, flows and dikes from Northeastern Paraná Magmatic Province (PMP), were submitted to paleomagnetic, chemical and radiometric analyses. The rocks are high in TiO_2 content, and similar in composition to the rocks from the northern region of PMP. The sills intrude mainly Paleozoic sediments, and can be subdivided into two domains; the northern being characterized by sills showing reversed polarities, and the southern essentially by sills of normal polarities. $^{40}Ar/^{39}Ar$ dating of three distinct sills gave plateau ages (129.9 ± 0.1, 130.3 ± 0.1 and 131.9 ± 0.4 Ma) that are similar to surface-outcropping flows of the Northern Paraná Basin, and the Ponta Grossa dikes. The new paleomagnetic data combined with existing data from the northern PMP allowed the calculation of a paleomagnetic pole at 71.4° E and 83.0° S ($N = 92$; $\alpha_{95} = 2.4°$; $k = 39$). This pole is in good agreement with poles for central and southern PMP, which are slightly older than the northern PMP, as well as for the contemporaneous Central Alkaline Province (Paraguay) on the western side of PMP. In contrast, the coeval pole for the Ponta Grossa dikes (eastern border of PMP), however, is slightly displaced from that

* Corresponding author. Tel.: +55-11-8184755; fax: +55-11-8185034.

E-mail addresses: marcia@iag.usp.br (M. Ernesto), irene@usp.br (M.I.B. Raposo), prenne@bgc.org (P.R. Renne), demin@univ.trieste.it (A. de Min)

[1] Present address: Instituto de Geociências, Universidade de São Paulo Rua do Lago, Cidade Universitária, 05508-900 São Paulo, Brazil.

0264-3707/99/$ - see front matter © 1999 Elsevier Science Ltd. All rights reserved.
PII: S0264-3707(99)00013-7

group of poles, suggesting that dikes in that area may have undergone some tectonic tilting. © 1999 Elsevier Science Ltd. All rights reserved.

1. Introduction

It has been proposed that intracontinental deformations occurred in South America and Africa plates as a consequence of the evolution of the South Atlantic ocean. Curie (1984) proposed a right-lateral transcontinental strike-slip movement across the Paraná Basin in continuation to the Walvis–São Paulo ridges and Conceição et al. (1988) proposed that this discontinuity runs along the NW trending Curitiba–Maringá shear zone along the Ponta Grossa Arch. Rapalini et al. (1993) found no intracontinental movements within South American plate affecting paleomagnetic data from Late Carboniferous to Middle-Late Jurassic, by comparing statistically the data from northern and southern South America. On the other hand, Chang et al. (1992) argue that those assumptions lack field evidence and suggest a model involving strain accomodation by rotation of South America relative to Africa, producing extension in a NE–SW direction, and injection of NW–SE oriented dikes.

The paleomagnetic poles calculated by Ernesto et al. (1990) for the Serra Geral Formation (Paraná Magmatic Province; PMP) do not support the hypothesis of movements along the tectonic lineaments in the Paraná Basin (Fig. 1) after volcanic emplacement. However, the paleopole for the northern area of the PMP is still poorly defined compared to the other two poles for southern and central PMP, allowing speculations about age differences or internal deformations. Evidence of tilting in the area of the Ponta Grossa dikes (Fig. 1) became clear with additional data from volcanics of the PMP and Ponta Grossa dikes (Renne et al., 1996a, 1996b).

This paper presents paleomagnetic data from the northeastern border of the basin which significantly improve the definition of the northern PMP pole. This is a unique area within the Paraná basin because it exposes a great number of sills which are not so frequent in other areas. The temporal relationships between these intrusive rocks and the lavas are not yet clearly known, although there are some indications of their being contemporaneous (Renne et al., 1997). Chemically the sills are analogous to the northern PMP lavas (Bellieni et al., 1984), and therefore may help to improve the northern PMP paleomagnetic pole.

2. Geological setting

The intracratonic sedimentary Paraná Basin (Fig. 1) is located in central-eastern Brazil, and covers an area of about 1,700,00 km^2. Sedimentation started during Early Paleozoic and formed up to 4,000 m of sediments. According to Almeida (1981) marine sediments were deposited during Devonian–Lower Carboniferous followed by marine and continental sediments (the Itararé Formation of the Tubarão Group) subjected to the Permian–Carboniferous glaciation. Sedimentation then changes to fluviatile and deltaic types, and subsequently to marine platform to coastal-swamp environments (Rio Bonito and Palermo Formations: Tubarão Group; Irati Formation: Passa Dois Group). The end of the Paleozoic is

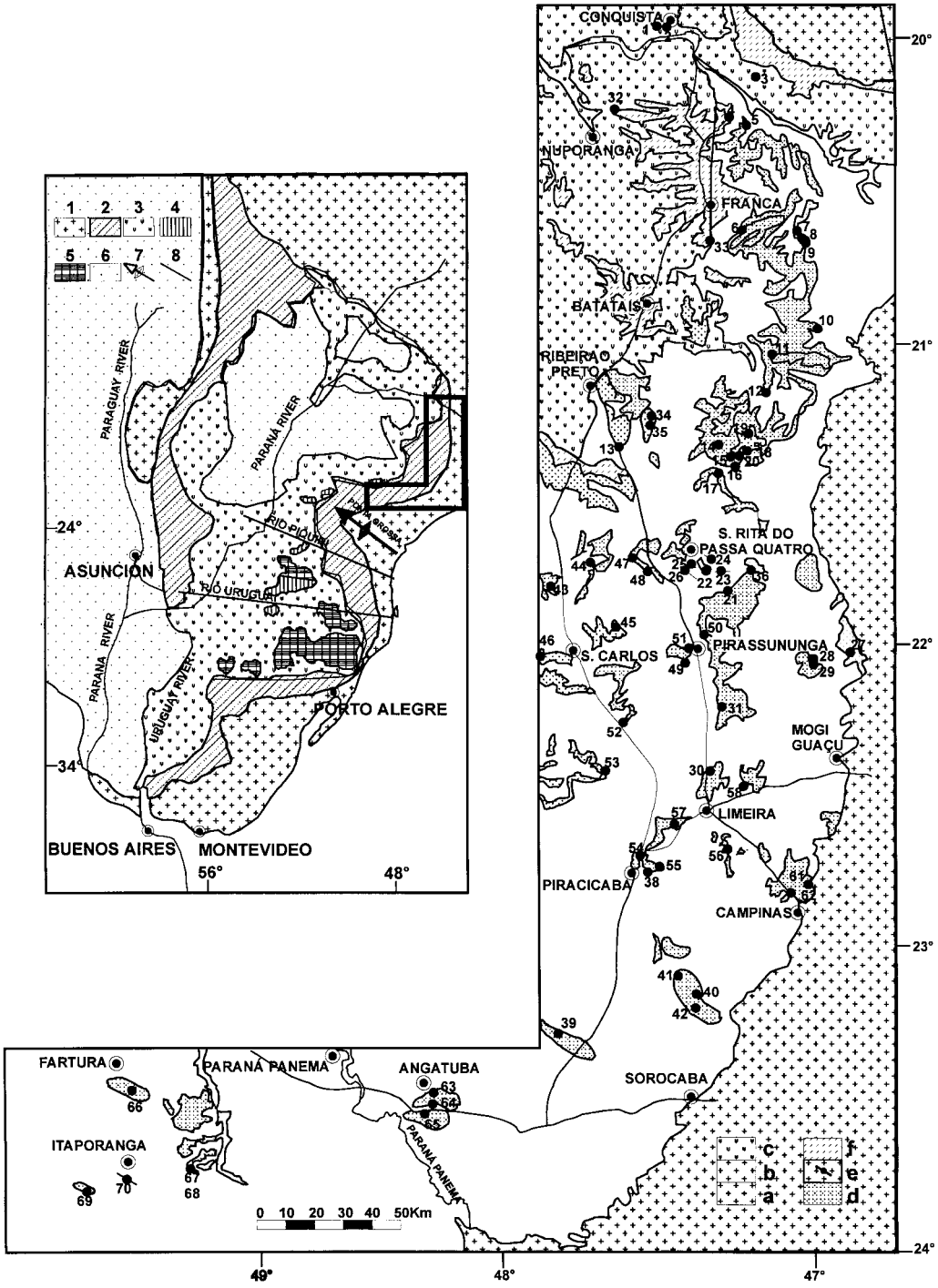


Fig. 1. Geological map of the Paraná Basin showing the sampling sites; (1) crystalline basement; (2) pre-volcanic sediments; (3) basic rocks; (4) Palmas acid rocks; (5) Chapecó acid rocks; (6) post-volcanic sediments; (7) arch-type structures; (8) structural lineaments. Study area: (a) crystalline basement; (b) pre-volcanic sediments; (c) flows; (d) sills; (e) dikes; (f) post-volcanic sediments.

characterized by fluviatile and deltaic sedimentation (Corumbataı and Rio do Rastro Formations: Passa Dois Group). At the end of Paleozoic until Jurassic, sedimentation was typically continental forming fluviatile sandstones and siltstones (Pirambóia Formation). At the beginning of the Jurassic desert conditions were prevalent and deposition of aeolian sandstones (Botucatu Formation) occurred in the entire basin. These sediments represent the top of the Gondwana sequence, immediately underlying, and in places interfingering with lavas of the Serra Geral Formation that covered the entire basin.

Although the Serra Geral volcanics generally overlie the Botucatu sandstones, they are locally found covering the Passa Dois Group. In the north-west border of the basin Petri and Fúlfaro (1983) reported lavas in contact with the pre-Devonian basement. In the northern part of the basin the volcanics are overlain by Upper Cretaceous sediments, mainly the Bauru Formation. The Serra Geral lavas are usually over a kilometre thick; the maximum observed thickness being about 1,700 m, at the drill hole of Cuiabá Paulista, State of São Paulo (Almeida, 1986). These rocks are predominantly tholeiitic and andesitic basalts, however, volcanic suites also include acid flows, which occur mostly towards the eastern fringes of the basin (e.g., Piccirillo et al., 1988).

Tectonic lineaments, e.g., the Rio Uruguai and Rio Piquiri lineaments in Fig. 1, cut the basin from its eastern edge to its centre. Some lineaments are characterized by aeromagnetic anomalies (Ferreira, 1982), others are represented by sets of faults. These tectonic features evolved since the Devonian and were particularly active during Triassic–Jurassic times (Fúlfaro et al., 1982). Subparallel to them are the axes of arch-type structures. The most outstanding of them, the Ponta Grossa arch, is also characterized by pronounced NW–SE magnetic anomalies (Ferreira, 1982), and is intruded by hundreds of diabase dikes trending preferentially subparallel to the arch axis (Raposo and Ernesto, 1995).

Geological and petrological data (Piccirillo et al., 1988) suggest a subdivision of the basin according to the distribution and nature of the different volcanic rock types. In the southern Paraná Basin (SPB; south of the Rio Uruguai lineament), basalts are characterized by low titanium content (<2 wt%), and by the existence of acid flows of Palmas type (rhyodacites and rhyolites) on the upper part of the sequences. The northern (NPB; north of the Rio Piquiri lineament) is dominated by tholeiites with high titanium content (>2 wt%) that are sometimes overlain by porphyritic rhyodacite flows (Chapecó type) towards the southeasternmost areas. In the central Paraná basin (CPB) the volcanic suites are similar to those occurring in the SPB and NPB, and may include both Palmas and Chapecó types. Here again acid flows tend to concentrate towards the eastern edge of the basin.

Recently, a large number of $^{40}Ar/^{39}Ar$ determinations for the Paraná rocks (Renne et al., 1992a, 1992b, 1996a, 1997; Turner et al., 1994) have become available. All these data show that only a few million years (~133–130 Ma) were required for the emplacement of the major part of the igneous rocks in the Paraná Basin. Detailed $^{40}Ar/^{39}Ar$ determinations (Renne et al., 1992a), including data from the bottom and top of the flow sequences, give a SPB mean age of 133 ± 1 Ma whereas, ages from the CPB and NPB concentrate around 132 Ma (Renne et al., 1992b, 1997), tending to be slightly younger than SPB. For the NPB, Turner et al. (1994) report ages varying from 137 ± 0.7 to 128.7 ± 1.1 Ma obtained mainly from borehole samples.

Diabase sills are found all over the basin intruding preferentially the Paleozoic sediments (Itararé and Irati Formations), and rarely the Botucatu sandstones or the Serra Geral flows

(Melfi and Girardi, 1963; Davino et al., 1982). The major volume of sills are found in the NPB in sections up to 1,000 m thick (Zalán et al., 1986). Sills outcropping in the northeastern fringe of the basin were described by several authors (e.g., Melfi and Girardi, 1963; Asmus, 1965; Wernick and Landim, 1966; Wernick and Ebert, 1967; Wernick, 1972; Soares et al., 1973; Davino et al., 1982). In general, sill thicknesses vary between 2 and 200 m. Large thickness variability is also observed within a sill. Eventually they show same characteristics (vesicles or fracture patterns) that are normally indicators of the top of flows. This is found in a sill near Irati, SP (Asmus, 1965), and in a sill near Guareı, intruding the Estrada Nova Group (Curvelo, 1940). The scarcity of outcrops precludes estimating the areal extent of a single sill, although Cordani and Vandoros (1967) suggest that the Piracicaba–Limeira (State of São Paulo) sill covers an area of 900 km^2.

Geochemical and petrological data (Bellieni et al., 1984) from sills sampled in boreholes widely spread over the basin, indicate that these rocks are of tholeiitic or transitional character, and correspond in composition to the basalts of the Serra Geral Formation, mainly the high titanium tholeiitic rocks from NPB.

3. Paleomagnetism

Samples were generally collected along road cuts. Most samples are cylinders taken with the aid of a gasoline-powered rock drill although some hand-blocks were also taken, mainly from the northernmost areas. Sample orientation was performed by both magnetic and solar compasses, whenever possible. Normally, 6–9 cylinders (or 3 blocks) were collected from each outcrop. In favorable outcrops more detailed sampling was performed. Sampling covered a wide area spanning south latitudes of nearly 20° and 24°. More than 500 samples (including blocks and cilynders) from 70 outcrops were sampled (Fig. 1).

For paleomagnetic measurements samples were cut into specimens of 2.5 cm diameter and 2.2 cm height. At least three specimens from each outcrop (site) were analyzed. Remanent magnetizations were measured using a Digico spinner magnetometer. Samples were submitted to both alternating field (a.c.) and thermal demagnetization. For a.c. cleaning both tumbling (Molspin Ltd.) and static (Sapphire Instruments) demagnetizers were used. In pilot-specimens a.c. cleaning was performed from fields of 5 mT up to 80 or 100 mT in steps of 5 mT. Some samples were thermally demagnetized from 150 to 500°C in steps of 50°C, and then, when necessary up to 580°C in steps of 20 or 30°C, using a Schonsdted furnace. An a.c. field of 10–15 mT was applied before the thermal cleaning in order to remove soft remanence components. Possible changes in susceptibility values were monitored using a Sapphire Instruments SI-1 susceptibility meter. The demagnetization data have been analyzed by means of Zijderveld plots (Zijderveld, 1967) and Wulff stereograms, as seen in Fig. 2. The majority of the samples revealed high magnetic stability and no strong low-stability magnetization was detected. Few samples showed a low-stability component of magnetization which was easily eliminated by fields not greater than 40 mT or temperatures up to 400°C.

Typical thermomagnetic curves (Fig. 3) for different rock types, and rocks from different areas, indicated that remanences are carried by Ti-poor titanomagnetites with Curie temperatures ranging from 500 to 580°C, but concentrating around 530–550°C. Under reflected

(a)

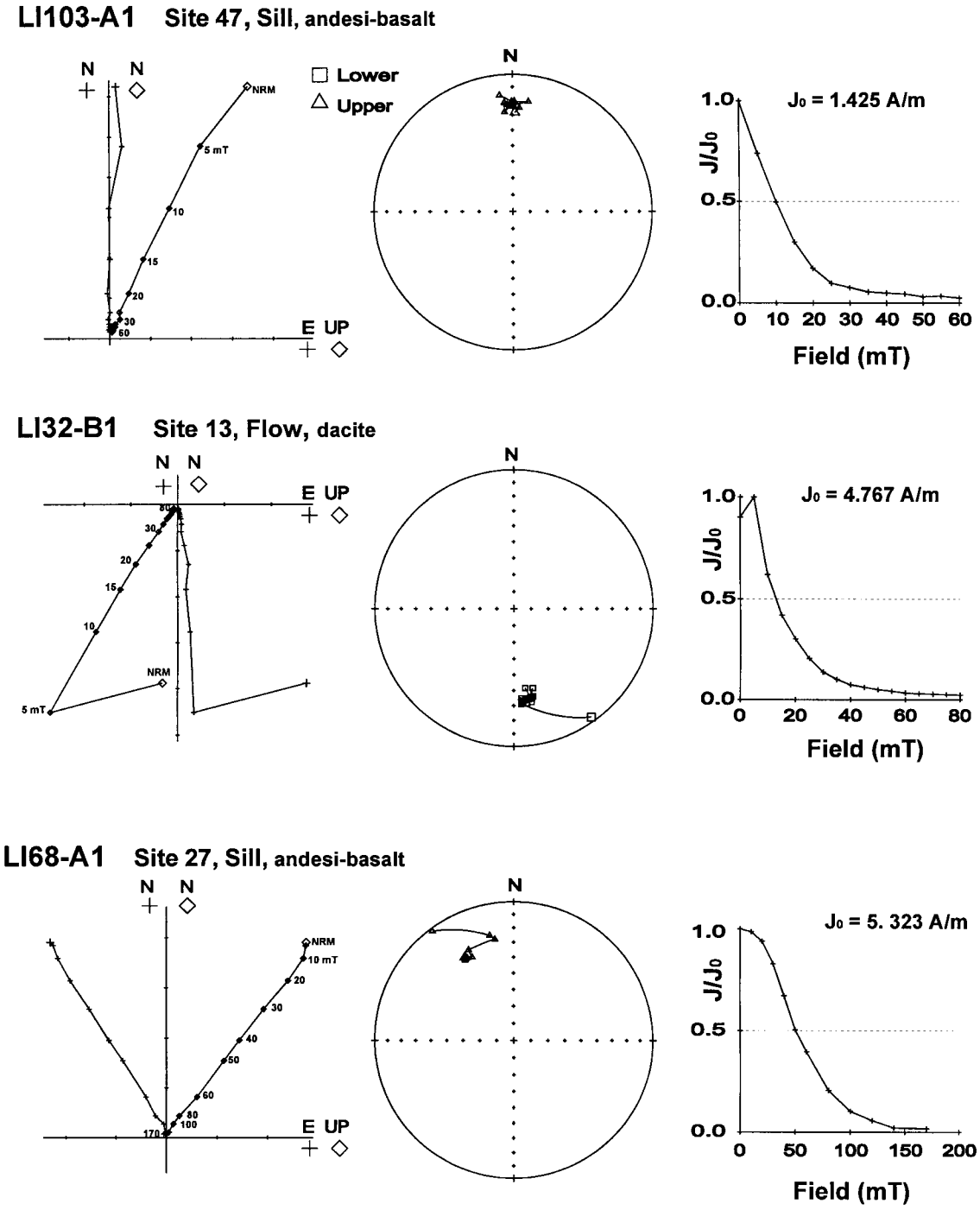


Fig. 2. Orthogonal vector plots, stereoplots, and magnetic intensity for demagnetization of samples from various sites and rock-types. Intensities are in SI units.

(b)

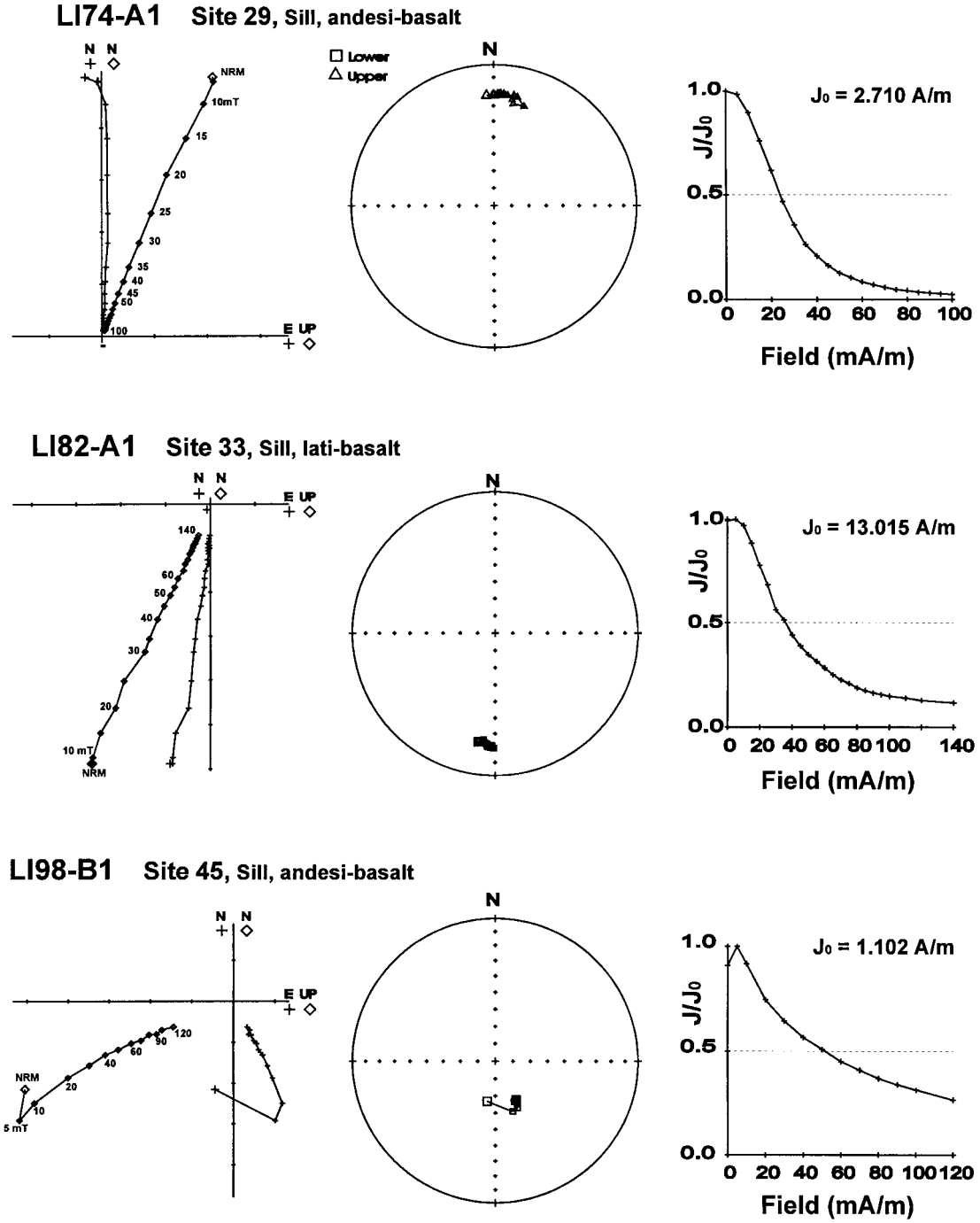


Fig. 2 (*continued*)

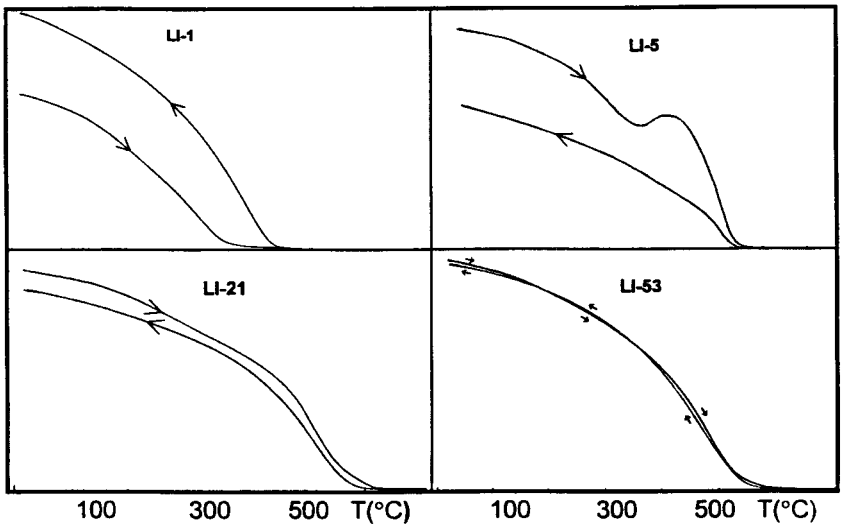


Fig. 3. Typical thermomagnetic curves for sills and flows. Heating rate was 50°C/min.

light phenocrysts and groundmass crystals of titanomagnetites show typical textures of high temperature oxidation like trellis, composite and sandwich (Haggerty, 1981), and phenocrysts and microphenocrysts of ilmenite.

The mean magnetization direction of each site (Table 1) was obtained by averaging directions from the corresponding specimens. Sites show normal and reversed polarities (Fig. 4(a)), and the

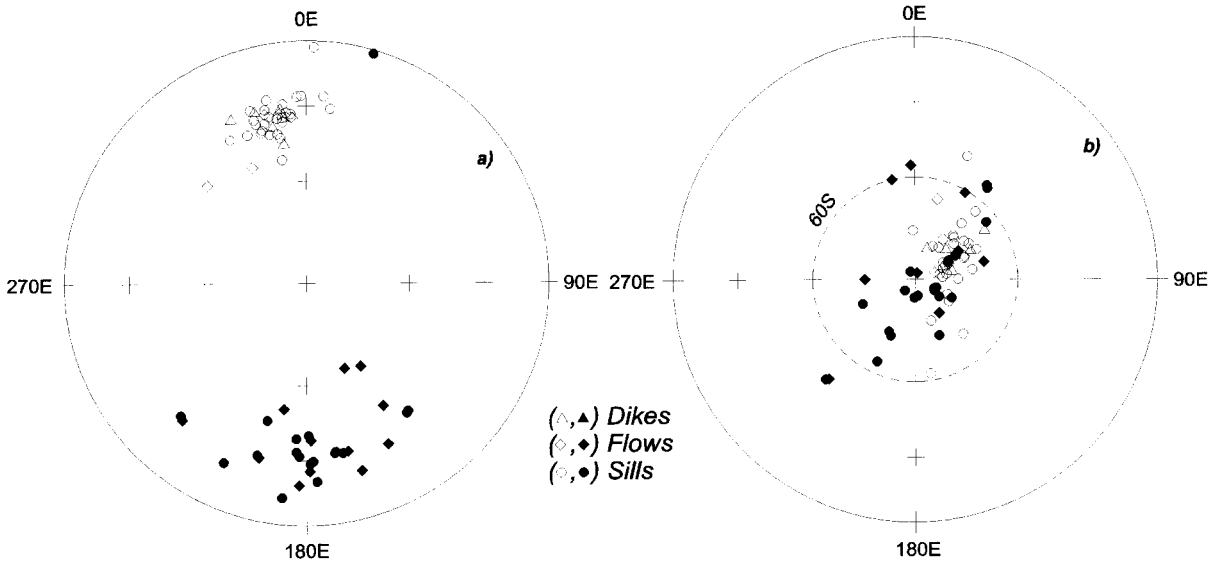


Fig. 4. (a) Stereographic plots for site means. Open symbols, negative inclinations; full symbols, positive inclinations; (b) South VGPs for sills, flows and dikes of the study area; open and full symbols indicate sites of normal and reversed polarities, respectively.

Table 1
Paleomagnetic data from Northeastern PMP rocks[a]

Site location					Mean magnetization direction							Virtual geomagnetic poles				
Code[b]	Longitude (°W)	Latitude (° S)	Altitude (m)	Type	n	$k \times 10^{-2}$ (SI)	Dec. (°)	Inc. (°)	α_{95} (°)	k	Pol.	Longitude (°E)	Latitude (°)	dp (°)	dm (°)	Chem. class.
1*	47°38′04″	19°57′13″	600	flow	8	1.60	222.4	26.7	3.9	201	R	221.0	−49.2	7.6	4.3	A–B
2*	47°16′37″	19°57′23″	450	flow	7	3.64	314.7	−46.1	27.1	7	N	203.1	48.1	51.8	37.9	LA
3	47°16′48″	20°07′12″	810	flow	10	1.13	182.2	18.7	8.4	34	R	144.5	−79.3	16.5	8.8	A–B
4	47°19′12″	20°13′11″	660	sill	4	1.63	195.5	28.6	10.6	77	R	206.6	−74.4	20.5	11.8	DAC
5	47°19′12″	20°15′00″	810	sill	9	5.53	182.6	31.7	3.5	372	R	171.6	−86.0	6.8	4.0	A–B
32*	47°46′09″	20°11′58″		sill	4	7.40	205.0	21.1	6.1	229	R	204.9	−64.2	11.9	6.4	L–B
6	47°10′12″	20°36′27″	985	sill	6	4.02	177.1	20.3	3.9	304	R	117.0	−79.5	7.6	4.1	L–A
7*	47°03′00″	20°36′47″	930	sill	12	5.81	223.7	27.6	6.9	41	R	222.1	−48.1	13.4	7.6	A–B
8	47°04′55″	20°37′37″	890	sill	13	5.27	178.9	28.5	1.4	856	R	120.9	−84.8	4.6	2.7	A–B
9	47°03′48″	20°38′12″	985	sill	6	3.77	178.5	29.2	2.4	614	R	115.8	−84.8	4.6	2.7	
33	47°26′36″	20°35′55″	780	sill	11	3.69	179.1	25.2	2.6	307	R	125.8	−82.6	5.1	2.8	A–B
10	47°00′00″	20°58′11″	480	sill	10	3.73	170.7	32.3	3.2	224	R	62.7	−80.6	6.2	3.7	AND
11	47°10′12″	21°03′00″	890	sill	9	5.11	177.9	29.6	4.3	74	R	111.7	−84.4	8.3	4.8	A–B
12	47°36′37″	20°57′22″	940	sill	7	5.09	168.0	31.8	4.6	175	R	59.4	−78.0	8.9	5.3	L–A
14	47°21′00″	21°18′00″	715	sill	4	8.75	153.2	28.8	4.7	386	R	51.4	−63.9	9.1	5.2	A–B
15	47°17′59″	21°19′47″	785	sill	8	1.83	183.8	33.7	3.3	278	R	183.0	−85.4	6.4	3.9	AND
16	47°19′47″	21°21′35″	750	sill	3	5.15	186.6	12.6	3.6	1204	R	156.7	−73.7	7.1	3.7	L–A
18	47°15′35″	21°18′00″	855	sill	7	2.06	196.1	44.8	15.4	16	R	244.7	−74.5	29.4	21.0	L–A
19	47°15′35″	21°16′48″	825	sill	9	3.90	179.4	40.7	8.5	38	R	328.2	−87.9	16.3	10.9	BT
13	47°43′11″	21°16′48″	670	flow	9	3.12	170.4	33.0	2.5	428	R	60.8	−80.3	4.8	2.9	DAC
34	47°34′47″	21°11′53″	690	flow	27	2.77	166.2	31.9	2.0	199	R	56.8	−76.4	3.9	2.3	AND
35	47°34′47″	21°11′53″	700	flow	26	2.11	178.3	38.6	2.0	196	R	191.1	−88.4	3.8	2.5	A–B
43	47°54′03″	21°47′18″		flow	3	2.78	190.3	50.5	5.5	505	R	270.0	−76.8	10.5	8.4	AND
44	47°55′47″	21°42′20″	630	flow	3	2.19	196.2	29.4	4.4	784	R	203.6	−73.5	8.5	4.9	A–B
45	47°38′50″	21°49′16″		flow	6	3.93	156.2	63.2	2.6	676	R	347.1	−60.1	5.0	5.7	A–B
46*	47°38′55″	2147′12″		flow	6	2.51	146.7	61.2	2.7	600	R	358.0	−55.6	5.2	5.5	
36*	47°12′30″	21°42′15″		sill	5	0.87	141.5	36.3	10.7	51	R	37.5	−54.1	20.5	12.9	
50	47°22′09″	21°55′06″		sill	3	2.21	142.4	35.8	6.6	351	R	38.5	−54.9	12.7	7.9	A–B
51	47°41′42″	22°09′53″	870	sill	4	422.17	184.1	39.2	5.8	255	R	222.2	−86.3	11.1	7.3	A–B
23	47°22′11″	21°43′11″	720	dike	3	2.93	334.4	−28.4	2.7	2015	N	234.2	65.8	5.2	3.0	L–A
24	47°22′48″	21°40′12″	865	dike	6	3.49	343.2	−28.9	4.2	253	N	241.2	72.9	8.1	4.7	A–B
25	47°25′47″	21°40′47″		dike	6	2.84	351.5	−30.0	11.0	38	N	255.9	80.2	21.3	12.4	AND
26	47°27′00″	21°41′59″	820	dike	6	1.79	347.9	−37.2	3.4	393	N	225.0	78.7	6.5	4.1	A–B
17	47°19′12″	21°22′47″	670	sill	6	6.76	2.1	−3.0	9.2	53	N	318.5	70.0	18.4	9.2	L–A
20	47°13′11″	21°21′00″	900	sill	7	6.99	16.5	1.8	5.6	119	N	350.8	63.5	11.2	5.6	A–B

(continued on next page)

Table 1 (*continued*)

Site location					Mean magnetization direction							Virtual geomagnetic poles				
21	47°19′12″	21°44′23″	880	sill	9	5.11	346.2	−40.2	6.0	74	N	214.9	77.2	11.5	7.6	A–B
22	47°25′12″	21°41′59″	755	sill	6	2.38	355.2	−34.9	6.2	119	N	250.5	84.9	11.9	7.4	A–B
47	47°38′58″	21°44′30″	627	sill	15	2.34	350.9	−32.7	3.7	106	N	245.5	80.6	7.1	4.3	A–B
48	47°33′41″	21°45′75″	627	sill	6	4.08	338.4	−37.9	5.4	157	N	219.8	69.9	10.4	6.7	L–A
27	46°52′48″	22°01′48″	740	sill	9	2.69	332.1	−36.4	4.6	127	N	221.9	64.0	8.8	5.6	A–B
28	47°01′48″	22°01′11″	630	sill	10	6.03	343.8	−37.9	5.5	78	N	222.8	74.9	10.5	6.8	A–B
29	47°01′12″	22°04′12″	710	sill	8	3.70	357.0	−25.7	4.2	177	N	294.6	81.0	8.2	4.6	A–B
37	47°02′59″	21°45′00″		sill	9	1.14	345.4	−35.4	3.7	198	N	227.4	76.2	7.1	4.4	
30	47°12′00″	22°21′00″	675	sill	9	5.11	346.9	−37.8	9.4	30	N	221.0	74.5	15.9	10.4	L–A
31	47°25′41″	22°12′04″	660	sill	3	12.50	351.8	−34.2	6.4	376	N	245.2	81.6	12.3	7.5	A–B
38	47°42′00″	22°41′59″		sill	6	0.88	351.6	−36.2	2.6	683	N	239.1	81.8	5.0	3.1	
49	47°28′25″	22°03′20″	792	sill	16	4.49	346.6	−29.6	4.0	87	N	246.3	75.9	7.7	4.5	A–B
52	47°42′05″	22°16′21″	840	sill	18	63.49	352.8	−34.1	2.4	200	N	248.9	82.4	4.6	2.8	AND
53	47°38′30″	22°23′05″	510	sill	9	2.61	349.9	−34.4	1.8	818	N	240.8	79.9	3.5	2.1	LAT
54	47°39′11″	22°43′01″		sill	9	3.42	347.4	−32.9	2.5	425	N	238.8	77.5	4.8	2.9	A–B
55	47°34′27″	22°47′39″		sill	9	5.94	342.5	−34.7	3.9	176	N	231.9	73.3	7.5	4.6	A–B
56	47°24′29″	22°45′55″	630	sill	12	3.49	342.0	−28.1	2.2	401	N	244.1	71.3	4.3	2.4	A–B
57	47°21′22″	22°36′53″	480	sill	11	3.38	354.7	−34.3	3.8	148	N	259.0	83.8	7.3	4.5	L–A
58	47°15′12″	22°29′23″	780	sill	9	4.37	353.3	−32.3	2.4	478	N	259.7	82.0	4.6	2.8	A–B
40	48°23′59″	23°12′00″		sill	6	0.64	148.0	43.9	6.5	107	R	30.1	−60.8	12.4	8.8	
41	48°30′00″	23°06′00″		sill	9	1.08	163.6	21.9	4.5	133	R	75.7	−70.5	8.8	4.8	
42	48°23′59″	23°11′59″		sill	7	0.48	335.2	−50.7	10.6	34	N	195.7	66.6	20.3	16.3	
59	46°57′15″	22°36′50″	630	sill	6	4.71	349.1	−40.8	4.0	286	N	216.8	79.9	7.7	5.1	A–B
60	46°57′48″	22°37′01″	600	sill	6	3.29	349.9	−42.7	2.5	739	N	207.9	80.5	4.8	3.3	BT
61	47°01′16″	22°45′33″	510	sill	9	4.41	3.54.7	−33.0	2.1	607	N	265.9	83.1	4.0	2.4	A–B
62	47°02′52″	22°50′33″	690	sill	9	3.78	342.2	−32.6	3.4	234	N	236.7	72.6	6.6	3.9	A–B
39	47°53′59″	23°18′00″		sill	15	0.70	346.2	−31.5	3.6	116	N	245.4	75.6	6.9	4.1	
63	48°23′01″	23°30′17″	594	sill	10	4.53	347.7	−25.5	3.9	156	N	260.4	74.6	7.6	4.2	
64	48°22′18″	23°31′11″	990	sill	6	10.24	358.4	−25.3	6.0	126	N	302.9	79.7	11.7	6.5	
65	48°24′45″	23°31′44″	858	sill	3	5.77	5.4	−25.3	3.9	985	N	339.1	78.6	7.6	4.2	
66	49°21′22″	23°27′50″	627	sill	8	3.23	349.0	−51.3	4.8	134	N	177.1	77.1	9.2	7.5	
67	49°21′32″	23°40′17″	480	sill	–	–	–	–	–	–	–	–	–	–	–	
68	49°21′35″	49°40′22″	490	dike	–	–	–	–	–	–	–	–	–	–	–	
69	49°31′52″	23°45′56″	495	sill	9	2.13	352.5	−28.6	2.5	427	N	269.5	78.9	4.8	2.8	
70	49°27′48″	23°44′21″	528	dike	10	2.17	350.9	−44.8	2.5	364	N	200.7	81.3	4.8	3.4	

[a] *n*, number of specimens included in the mean; Dec. and Inc., declination and inclination; α_{95} and *k*, Fisher (1953) statistical parameters; Pol., polarity (N = normal, R = reversed),; Long. and Lat., longitude and latitude; dp and dm, 95% confidence oval. Chem. class., chemical classification according to de La Roche et al. (1980); A–B = andesi–basalt; AND = andesite; L-A = lati-andesite; L-B = lati-basalt; DAC = dacite; BT = tholeiitic basalt; LAT = latite.

[b] Site numbers with an asterisk indicate data not included in the calculation of the paleomagnetic pole.

corresponding virtual geomagnetic poles (VGPs; Fig. 4(b)) show no distinction between sills, flows or dikes.

4. Geochemistry

Samples from 52 sites studied in this paper were submitted to bulk analyses of major, minor and trace elements by means of X-ray fluorescence. The analyses were carried out at the University of Trieste (Italy) following the procedures described by Bellieni et al. (1983). Analytical precision is 5% for the major and minor oxides, and better than 10% for the trace elements. These rocks are high in TiO_2 content, and according to de La Roche et al. (1980) they are classified as tholeiitic basalts, tholeiitic andesi-basalts, tholeiitic andesites, lati-andesites and dacites (Fig. 5; Table 1). Data from sills, flows and dikes plot in the same fields. The two samples that are distinguishable from the main group by their alkaline affinity, plot in the field of lati-basalt (site 32) and latite (site 53). The former has a strong potassic character (K_2O =

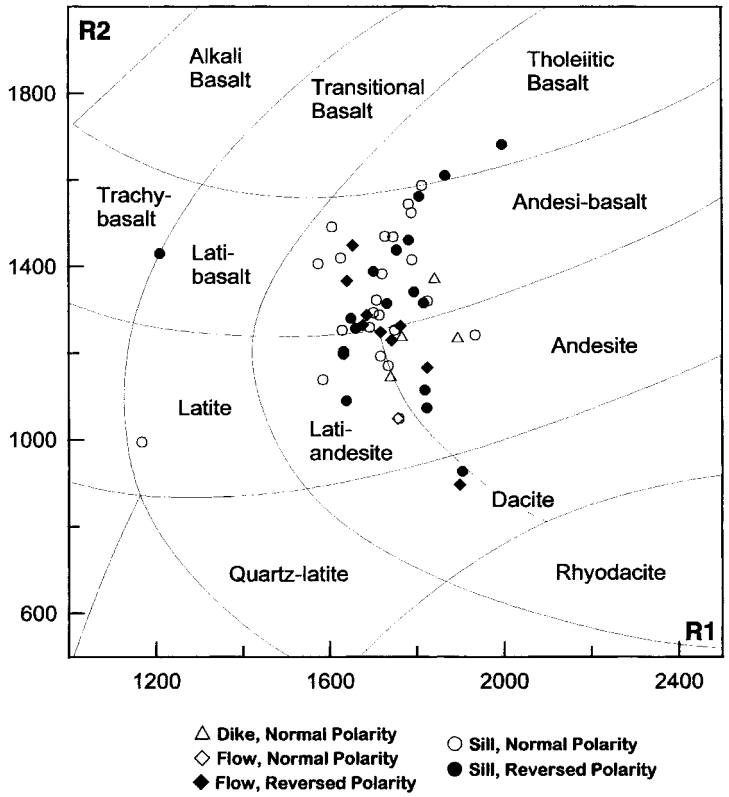


Fig. 5. Distribution of sills (circles), flows (diamonds) and dikes (triangles) in the classification diagram of de La Roche et al. (1980) modified by Bellieni et al. (1981). Open symbols: normal polarity group; full symbols: reversed polarity group.

5.56, Na_2O = 1.05 and MgO = 7.80 wt%; not shown in Fig. 6) and very likely is not related to the Early Cretaceous Paraná magmatism. Sample from site 53 falls in the latite field, probably due to alteration, therefore it was not considered in the geochemical study.

The variation diagrams of selected major and trace elements (Fig. 6), using MgO as a magmatic evolution index, show a strong similarity between sills, flows and dikes, independently of normal or reversed polarities. These geochemical characteristics are very close to those shown by borehole samples of NPB sills (Bellieni et al., 1984) and by high-TiO_2 NPB

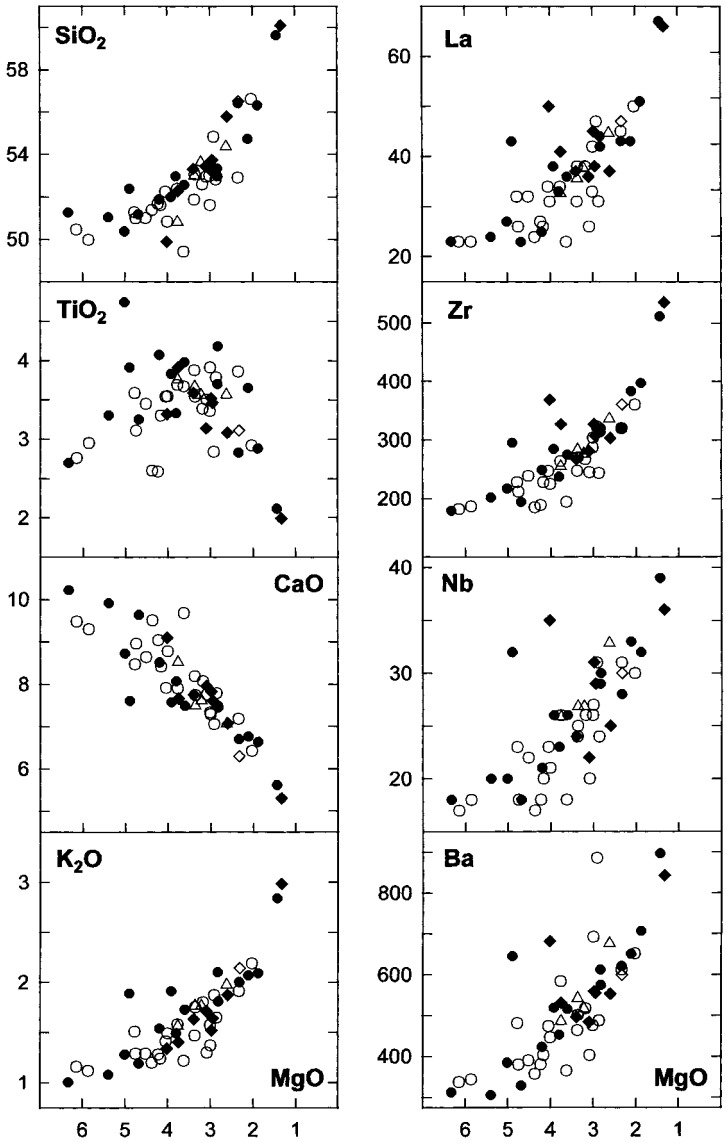


Fig. 6. Selected major (wt%) and trace (ppm) elements versus MgO (wt%). Symbols as in Fig. 5.

flows (Piccirillo et al., 1988), indicating similar parental magmas. The significant variations in major, minor and trace elements indicate that differentiation is compatible with low-pressure (gabbro) fractional crystallization, which in the thickest sills mainly occurred *in situ*.

5. $^{40}Ar/^{39}Ar$ Geochronology

For $^{40}Ar/^{39}Ar$ dating, fresh pure plagioclase separates of ca. 30 mg were prepared from

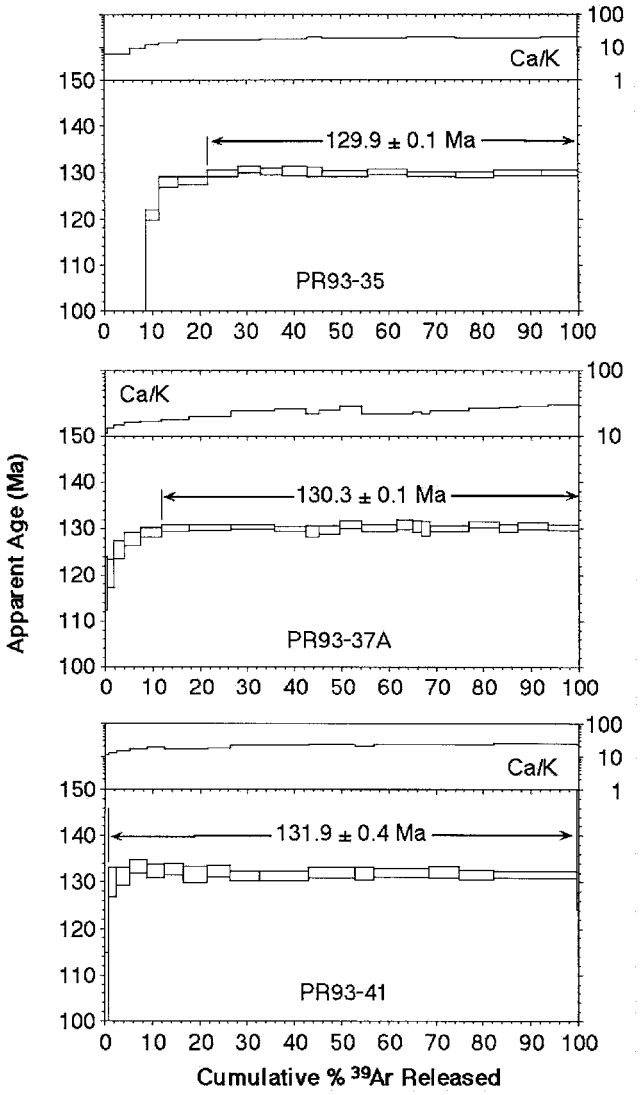


Fig. 7. $^{40}Ar/^{39}Ar$ apparent age and Ca/K spectra for three samples indicated. Age errors for steps are shown at 2s, but plateau age errors are indicated at 1s. Errors on plateau ages do not include uncertainties in age of standard or decay constants.

samples PR93-35, PR93-37A, and PR93-41. These samples correspond to sites 2, 9 and 31 (Table 1), respectively. Sample preparation, irradiation, and analysis followed methods and facilities described by Renne et al. (1996a) for the argon-ion laser-heating system at the Berkeley Geochronology Center. Neutron fluence was monitored with the Fish Canyon sanidine (FCs), using an age of 27.84 Ma for this standard. Recent studies (e.g., Renne et al., 1998) have shown that a more correct age for this standard is 28.02 ± 0.16 Ma (1s error excluding decay constant uncertainty), but the previous age is used herein to facilitate comparison with previous $^{40}Ar/^{39}Ar$ data from the Paraná–Etendeka Province, which are all based on 27.84 Ma for FCs or (equivalently) 520.4 Ma for the MMhb-1 standard.

All three samples yield discordant age spectra (Fig. 7), with anomalously young apparent ages at low temperatures, but all three nonetheless yield well-defined plateaux corresponding to $>78\%$ of the ^{39}Ar released. In all three cases, the discordance is correlated with anomalously low Ca/K, suggesting that the low- temperature effects are associated with subtle alteration products such as sericite. Plateau values of Ca/K are uniform, indicating that the plateau ages are unlikely to be corrupted by the secondary effects. The plateau ages of PR93-35 (129.9 ± 0.1 Ma) and PR93-37A (130.3 ± 0.1 Ma) are nearly indistinguishable, but that of PR93-41 (131.9 ± 0.4 Ma) is slightly older.

These results are similar to those obtained for the Ponta Grossa dikes (Renne et al., 1996a) and surface-outcropping flows of the Northern Paraná Basin (e.g., Renne et al., 1997), but discernibly younger than the vast majority of $^{40}Ar/^{39}Ar$ ages from flows in the Southern Paraná Basin (Renne et al., 1992a, 1992b, 1997; Turner et al., 1994) or the adjacent Etendeka Province of Namibia (Renne et al., 1996b). These data thus provide additional evidence supporting a northward migration of magmatism during peak activity of the province, which apparently lasted from 133 to 130 Ma.

6. Discussion

Paleomagnetic results are given in Table 1, organized in order of ascending latitudes. Sites that may belong to the same unit (flow or sill) or that intrude the same stratigraphic levels (bottom, middle or top of the Paleozoic sequence) are grouped. The study area can be clearly subdivided into two polarity zones; the northern zone (up to latitude $\sim 21^{\circ}30'$ S) is dominated by reversed polarities, whereas normal polarities characterize the southern zone. Both zones are chemically similar suggesting that the magmatic activity in the area was restricted to two main pulses controlled by a migrating tectonic regime. Paleomagnetic data are also very consistent, indicating that the time elapsed during the acquisition of thermal remanence was very short. This is especially true for the normal polarity sites for which mean directions are well grouped indicating low between-site dispersion (Fig. 4(a)). Reversed directions are more scattered than normal directions. Although this scattering could be attributed to the influence of the present geomagnetic field, it may also reflects the unstable behavior of the geomagnetic field during Early Cretaceous times due to the high reversal frequency. The reversed chrons were in general of shorter duration causing the paleomagnetic record to be more dispersed during those events. Intermediate directions (sites 1, 2, 7, 36, 46 and 50) associated to VGPs located more than 30°

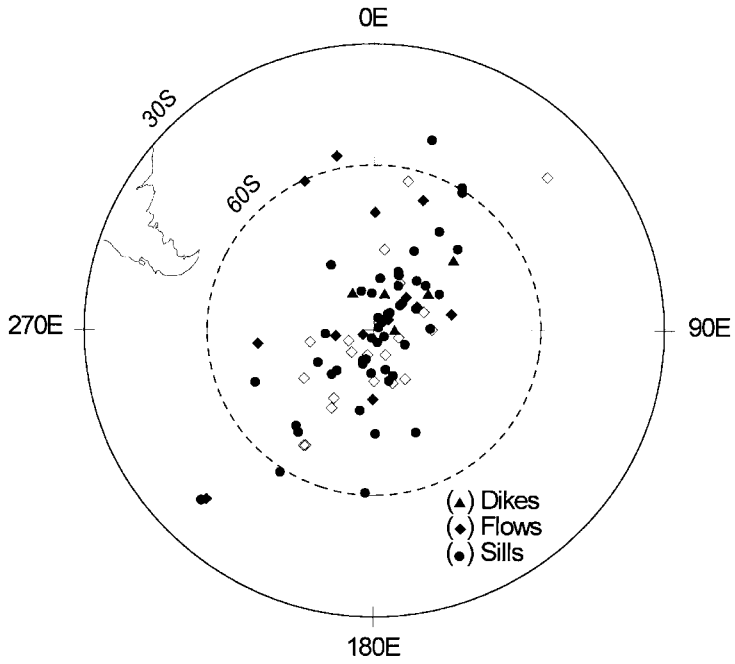


Fig. 8. VGPs for flows, sills and dikes studied in this paper, along with other existing data (open symbols) from NPB (Ernesto et al., 1990).

apart from the mean (Fig. 4(b)), probably related to polarity transitions or geomagnetic excursions, generally have positive inclinations, indicating reversed polarities.

The new paleomagnetic data from the northeastern border of the basin give a similar VGP

Table 2
Paleomagnetic poles for the Southern, Central and Northern Paraná Basin and age-related magmatism surrouding the basin[a]

Code	Age (Ma)	N	Longitude	Latitude	α_{95}	k	After correction		Reference
			(° E)	(° S)	(°)		(° E)	(° S)	
SPB	~ 133[b]	197	106.2	84.0	1.5	45	87.5	84.9	Raposo and Ernesto (1995)
CPB	~ 132[c]	103	64.4	84.1	2.3	38	65.7	86.1	Raposo and Ernesto (1995)
NPB	~ 132[d]	92	71.4	83.0	2.4	39	57.1	85.8	This paper
PG	~ 130[e]	115	30.3	82.4	2.0	44	57.0	85.1	Raposo and Ernesto (1995)
CAP	127–130[f]	75	62.3	85.4	3.1	29	62.3	85.4	Ernesto et al. (1996)

[a] N, number of sites; α_{95} and k, Fisher (1953) statistical parameters.
[b] Renne et al. (1992a).
[c] Renne et al. (1992b).
[d] Renne et al. (1997).
[e] Renne et al. (1996a), Renne et al. (1996b).
[f] Velázquez et al. (1992).

distribution as data (mainly flows; Ernesto et al., 1990) from other areas of the NPB (Fig. 8). Combining all the existing results, a new paleomagnetic pole for NPB is located at 71.4° E 83.0° S ($N = 92$; $\alpha_{95} = 2.4°$; $k = 39$). The new NPB pole is not statiscally distinguishable from the CPB pole (Table 2; Fig. 9), as is expected since the rocks from these two areas are of the same age (Renne et al., 1992b, 1997). The circles of confidence of NPB and CPB poles, however, intercept only slightly the SPB confidence circle (Table 2; Fig. 9), suggesting that the poles may be distinct. This difference, however, cannot be attributed to apparent polar wander, as a time interval of about 1–3 million year (133–130 Ma) is not sufficient to produce continental drift large enough to be detected paleomagnetically. Therefore, there must be alternative explanations for the apparent non-coincidence of the poles.

First of all it is important to stress that the data set for the SPB includes almost 200 data from individual flows, and therefore is a more complete data set than those for CPB and NPB which are each nearly half the size of the SPB set. As a consequence, the VGP distribution for SPB is closer to the expected Fisher distribution than the distributions for CPB and NPB, as seen in Fig. 8. In this case the difference in pole positions might be attributed to the uncomplete removal of the secular variation effects for the latter poles. Another possibility to be considered is a tectonic correction for all the PMP data, since the lavas are essentially sub-horizontal, dipping gently (less than 5°) towards the center of the basin (e.g. Melfi et al., 1988). A rotation of 3° would bring all the three poles closer together (Table 2). However, since the differences are neglegible, and the corrections are to some extent arbitrary, it is preferable to consider these poles without corrections.

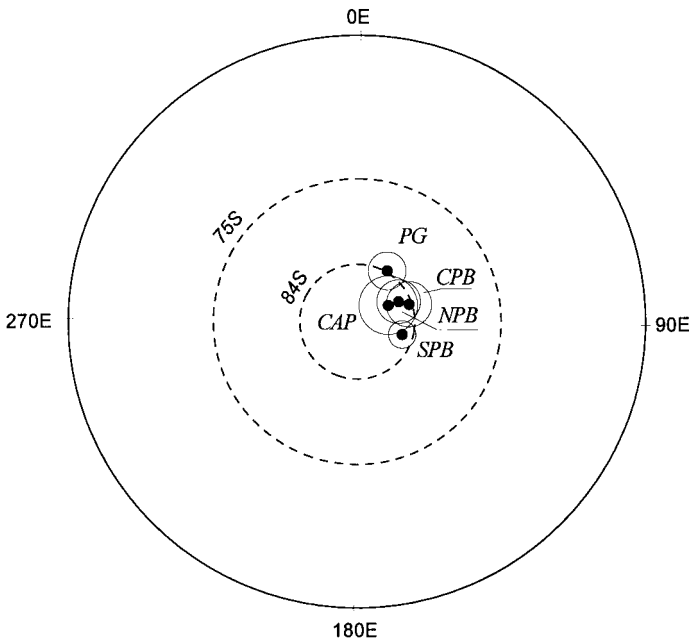


Fig. 9. Paleomagnetic poles for SPB, CPB and NPB along with poles for the Ponta Grossa dikes and the Central Alkaline Province in Paraguay. Open circles represent the 95% confidence circles.

Similar observations must be made regarding the Ponta Grossa pole. It differs statistically from the PMP mean pole, although their ages differ by only 2 or 3 million years. (main age of 130.5 Ma with minor activity in the 121–126 Ma interval; Renne et al., 1996a). The angular discordance between the two poles is equal to 6.8°, which if due to continental drift, would imply plate velocities greater than 20 cm/year. On the other hand, the paleomagnetic pole for the Central Alkaline Province (CAP) in Paraguay (Ernesto et al., 1996), which probably span the same time interval (127–130 Ma) as the Ponta Grossa dikes (based on K/Ar dating; Velázquez et al., 1992) is closer to the PMP poles (angular distance of ~2.7° implies more reasonable velocities). This observation reinforces the hypothesis of a tectonic influence on the PG pole.

Gimenez (1998) mapping the Ponta Grossa dikes in the northern sector of the Ponta Grossa Arch, found that the majority of the NW dikes dip subvertically (dips steeper than 80°) to S217-265W. Similar observations were done during sampling for the paleomagnetic work. Therefore, a tectonic correction for the Ponta Grossa paleomagnetic pole is plausible. This pole comes closer to the PMP and CAP poles when a net rotation of 5° is applied to an axis at N130E, which is approximately the orientation of the Ponta Grossa arch itself (Fig. 1). The rotated pole will be at 58.6° E and 84.4° S.

7. Concluding remarks

1. Flows and sills in northeastern Paraná Magmatic Province are contemporaneous, and are similar in chemical composition.
2. The rocks are high in TiO_2 content, and are classified as tholeiitic basalts, tholeiitic andesi-basalts, tholeiitic andesites, lati-andesites and dacites, according to de La Roche et al. (1980) classificative diagram. They are equivalent to other rocks from northern PMP.
3. $^{40}Ar/^{39}Ar$ dating of pure plagioclase separates from sites LI-2, LI-9, and LI-31 gave plateau ages of 129.9 ± 0.1, 130.3 ± 0.1 and 131.9 ± 0.4 Ma, respectively. These results are similar to those obtained for the Ponta Grossa dikes and surface-outropping flows of the Northern Paraná Basin, but discernibly younger than the vast majority of $^{40}Ar/^{39}Ar$ ages from flows in the Southern Paraná Basin or the adjacent Etendeka Province in Namibia, providing additional evidence supporting a northward migration of magmatism.
4. Sills cutting mainly Paleozoic sediments are the most frequent mode of occurrence of igneous rocks in the area, and may have been intruded during two main magmatic pulses. Reversed polarity sills characterize the northern part of the studied area, while normal polarity sills are dominant in the southern part.
5. Lateral extension of an individual sill cannot be assessed by means of paleomagnetic or geochemical data due to the homogeneity of data from site to site.
6. The VGPs for the analysed rocks are very consistent with those from other northern PMP areas, and the combined data allowed the calculation of a paleomagnetic pole located at at 71.4° E 83.0° S ($N = 92$; $\alpha_{95} = 2.4°$; $k = 39$).
7. This pole is in good agreement with poles from central and southern PMP, as well as for the contemporaneous Central Alkaline Province (Paraguay) on the western side of PMP. The

pole for the Ponta Grossa dikes is slightly displaced from that group of poles, suggesting that a tectonic tilting may have affected the dike remanences.

Acknowledgements

This project was supported by grants from CNPq, FAPESP, PADCT/FINEP, and NSF (EAR-9628065). The authors wish to thank E.M. Piccirillo for fruitful discussions, and J. Glen for a critical review. One of the authors (L.S.M.) is indebted to M.L. Detto for the decisive encouragement. Thanks are also due to Tamrat Endale for his help preparing diagrams.

References

Almeida, F.F.M., 1981. Sıntese sobre a tectônica da bacia do Paraná. In: Atas do II Simpósio Regional de Geologia, SBG, São Paulo.

Almeida, F.F.M., 1986. Distribuição regional e relações tectônicas do magmatismo pós-Paleozóico no Brasil. Revista Brasileira de Geocincias 16, 325–349.

Asmus, H.E., 1965. Notıcia preliminar sobre a ocorrência de estrutura amigdaloidal em sill de diabásio. In: Proceedings of XX Congresso Brasileiro de Geologia, 124–130.

Bellieni, G., Piccirillo, E.M., Zanettin, B., 1981. Classification and nomenclature of basalts. IUGS Subcomission on the Systematics of Igneous rocks. Circular 34, Cambridge, Contribution, 87, 1–19.

Bellieni, G., Brotzu, P., Comin-Chiaramonti, P., Ernesto, M., Melfi, A.J., Pacca, I.G., Piccirillo, E.M., Stolfa, D., 1983. Petrological and paleomagnetic data on the plateau basalt to rhyolite sequences of the southern Paraná basin (Brazil). Anais da Academia Brasileira de Ciências 55, 355–383.

Bellieni, G., Comin-Chiaramonti, P., Marques, L.S., Melfi, A.J., Piccirillo, E.M., Stolfa, D., 1984. Low-pressure evolution of basalt sills from bore-holes in the Paraná Basin Brazil. Tschermarks Mineralogie Petr. Mitt 33, 25–47.

Chang, H.K., Kowsmann, R.O., Figueiredo, A.M.F., Bender, A.A., 1992. Tectonics and stratigraphy of the East Brazil rift system: an overview. Tectonophysics 213, 97–138.

Conceiçáo, J.C., Zalan, P.V., Wolff, S., 1988. Mecanismo, evolução e cronologia do rift Sul-Atlântico. Boletim de Geocincias da Petrobrás 2, 255–265.

Cordani, U.G., Vandoros, 1967. Basaltic rocks of the Paraná basin. Problems in Brazilian Gondwana Geology, Paraná, pp. 207–223.

Curie, D., 1984. Overture de l'Atlantique Sud et discontinuités intraplaques: Une nouvelle analyse. Ph.D. thesis, Université Bretagne Occidentale, France, p. 192.

Curvelo, W. da S., 1940. Sobre um diabásio amigdalóide de Guareı São Paulo. Mineralogia e Metalogênese 10, 263–266.

Davino, A., Sinelli, O., Souza, A., Correia, C.T., 1982. Diabásios na região nordeste da Bacia do Paraná. Proceedings of the Thirty Second Congresso Brasileiro de Geologia, Salvado 4, pp. 1736–1744.

de La Roche, H., Leterrier, J., Grandclaude, P., Marchal, M., 1980. A classification of volcanic and plutonic rocks using R1–R2 diagram and major element analyses. Its relationships with current nomenclature. Chemical Geology 29, 183–210.

Ernesto, M., Pacca, I.G., Hiodo, F.Y., Nardy, A.J.R., 1990. Palaeomagnetism of the Mesozoic Serra Geral Formation, Southern Brazil. Physics of the Earth and Planetary Interiors 64, 153–175.

Ernesto, M., Comin-Chiaramonti, P., Gomes, C.B., Castillo, A.M.C., Velazquez, J.C., 1996. Paleomagnetic data from the Central Alkaline Province, Eastern Paraguay. In: Gomes, C.B., Comin-Chiaramonti, P. (Eds.), Alkaline Magmatism in Central-Eastern Paraguay. EDUSP/FAPESP, São Paulo, pp. 85–102.

Ferreira, F.J.F., 1982. Alinhamentos estruturais magnéticos da região centro- oriental da bacia do Paraná e seu sig-

nificado tectônico, Geologia da Bacia do Paraná, 12. Instituto de Pesquisas Tecnológicas, São Paulo, pp. 143–166.

Fisher, R.A., 1953. Dispersion on a sphere. Proceedings of the Royal Society of London Serie A 217, 295–305.

Fúlfaro, V.J., Saad, A.R., Santos, M.V., Vianna, R.B., 1982. Interpretação do tectonismo cratônico - o craton brasileiro e a bacia sedimentar do Paraná. Revista Brasileira de Geociências 12, 590–611.

Gimenez, D.Z., 1998. Caraterização tectono-estrutural do enxame de diques cretácicos na região de Fartura, Estado de São Paulo. M.Sc. thesis, University of the State of São Paulo, Rio Claro.

Haggerty, S.E., 1981. Oxidation of opaque mineral oxides in basalts. In: Rumble, D. (Ed.), Reviews in Mineralogy, Oxide Minerals, vol. 3. Mineralogical Society of America, Washington D.C (Chapter 4).

Melfi, A.J., Girardi, V.A.V., 1963. Ocorrência de um sill de diabásio no arenito Botucatu, Municıpio de Igarapava (SP). Boletim da Sociedade Brasileira de Geologia II (2), 55–70.

Melfi, A.J., Piccirillo, E.M., Nardy, A.J.R., 1988. Geological and magmatic aspects of the Paraná Basin - an introduction. In: Piccirillo, E.M., Melfi, A.J. (Eds.), The Mesozoic Flood Volcanism of the Paraná Basin: Petrogenetic and Geophysical Aspects. Universidade de São Paulo, São Paulo, pp. 1–13.

Petri, S., Fúlfaro, J.V., 1983. Geologia do Brasil. T.A. Queiroz e Universidade de São Paulo.

Piccirillo, E.M., Comin-Chiaramonti, P., Bellieni, G., Civetta, L., Marques, L.S., Melfi, A.J., Petrini, R., Raposo, M.I.B., Stolfa, D., 1988. Petrogenetic aspects of continental flood basalt-rhyolite suites from the Paraná Basin (Brazil). In: Piccirillo, E.M., Melfi, A.J. (Eds.), The Mesozoic Flood Volcanism of the Paraná Basin: Petrogenetic and Geophysical Aspects. Universidade de São Paulo, São Paulo, pp. 179–205.

Rapalini, A.E., Abdeldayem, A.L., Tarling, D.H., 1993. Intracontinental movements in western Gondwanaland: A palaeomagnetic test. Tectonophysics 220, 127–139.

Raposo, M.I.B., Ernesto, M., 1995. An Early Cretaceous paleomagnetic pole from Ponta Grossa dykes (Brazil): implications for the South America Mesozoic APWP. Journal of Geophysical Research 100 (B10), 20,095–20,109.

Renne, P.R., Ernesto, M., Pacca, I.G., Coe, R.S., Glen, J., Prévot, M., Perrin, M., 1992a. Rapid eruption of the Paraná flood volcanics, rifting of southern Gondwanaland and the Jurassic-Cretaceous boundary. Science 258, 975–979.

Renne, P.R., Ernesto, M., Pacca, I.G., Nardy, A.J.R., Coe, R.S., Glen, J.M., Prévot, M., Perrin, M., 1992b. Age and duration of Paraná flood volcanism in Brazil, EOS Transactions of the American Geophysical Union, 531.

Renne, P.R., Deckart, K., Ernesto, M., Féraud, G., Piccirillo, E.M., 1996a. Age of the Ponta Grossa dike swarm (Brazil), and implications to Paraná flood volcanism. Earth and Planetary Science Letters 144, 199–211.

Renne, P.R., Glen, J.M., Milner, S.C., Duncan, A.R., 1996b. Age of Etendeka flood volcanism and associated intrusions in southwestern Africa. Geology 24, 659–662.

Renne, P.R., Ernesto, M., Milner, S.C., 1997. Geochronology of the Paraná–Etendeka Magmatic Province. EOS Transactions of the American Geophysical Union, F742.

Renne, P.R., Swisher, C.C., Deino, A.L., Karner, D.B., Owens, T., DePaolo, D.J., 1998. Intercalibration of Standards, Absolute Ages and Uncertainties in $^{40}Ar/^{39}Ar$ Dating: Chemical Geology. Isotope Geoscience Section 145 (1–2), 117–152.

Soares, P.C., Sinelli, O., Penalva, F., Wernick, E., Souza, A., Castro, P.R.M., 1973. Geologia do Nordeste do Estado de São Paulo. Anais do XXVII Congresso Brasileiro de Geologia 27, 209–228.

Turner, S., Regelous, M., Kelley, S., Hawkesworth, C., Mantovani, M., 1994. Magmatism and continental break-up in the South Atlantic: high precision ^{40}Ar-^{39}Ar geochronology. Earth and Planetary Science Letters 121, 333–348.

Velázquez, V.F., Gomes, C.B., Capaldi, G., Comin-Chiaramonti, P., Ernesto, M., Kawashita, K., Petrini, R., Piccirillo, E.M., 1992. Magmatismo alcalino Mesozóico na porção centro-oriental do Paraguai: aspectos geocronológicos. Geochimica Brasiliensis 6, 23–35.

Wernick, E., 1972. Contribuição ao conhecimento da tectônica das rochas básicas da região nordeste do Estado de São Paulo. Notas Geomorfológicas 12, 13–30.

Wernick, E., Ebert, H., 1967. Novas observações sobre rochas básicas nos arredores de Rio Claro. In: Proceedings of the Twenty First Congresso Brasileiro de Geologia, São Paulo, 206–213.

Wernick, E., Landim, P.M.B., 1966. Nota sobre um diabásio amigdaloidal em Rio Claro (SP). Boletim da Sociedade Brasileira de Geologia 15, 61–66.

Zalán, P.V., Conceiçáo, J.C., Wolff, S., Astolfi, M.A., Vieira, I.S., Appi, V.T., Neto, E.V.S., Cerqueira, J.R.,

Zanotto, O.A., Paumer, M.L., Marques, A., 1986. Análise da Bacia do Paraná. Relatório Interno Gt-Os-009/85, Petrobrás, Rio de Janeiro.

Zijderveld, J.D.A., 1967. A.C. demagnetization of rocks: analysis of results. In: Collinson, D.W., Creer, K.M., Runcorn, S.K. (Eds.), Methods in Paleomagnetism. Elsevier, Amsterdam, pp. 254–286.